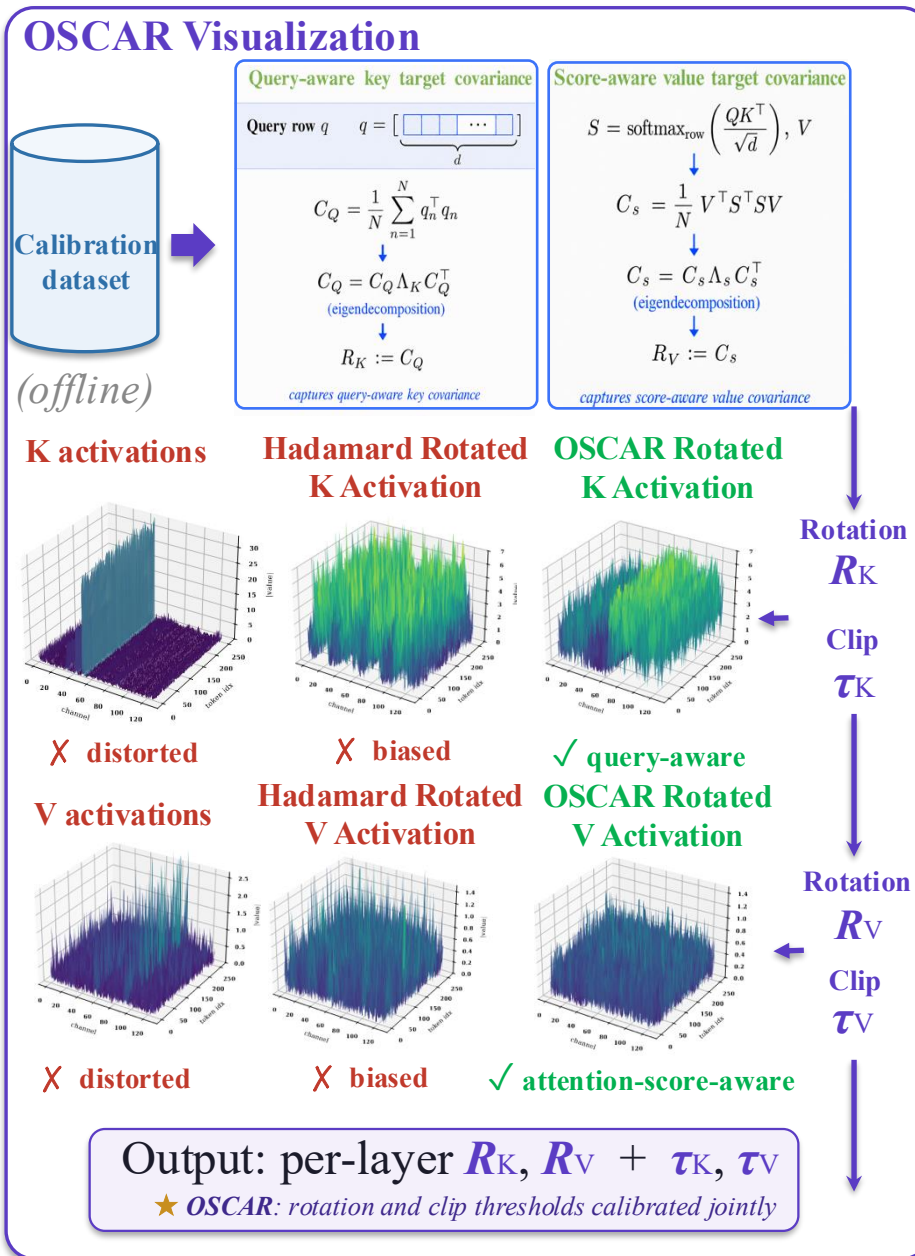
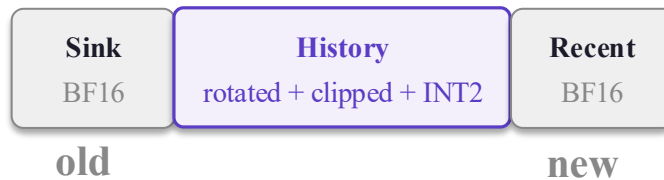




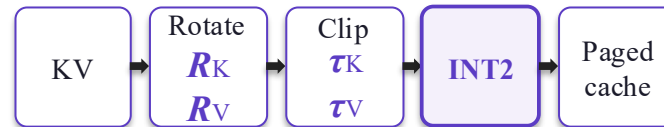
Inference

KV cache layout

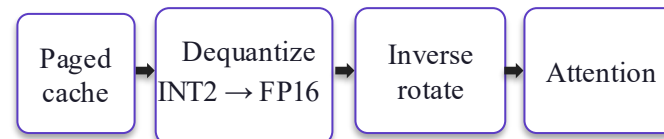


★ OSCAR keeps sink/recent in full precision

Write path



Read path



⚙️ **Fused Triton kernel**

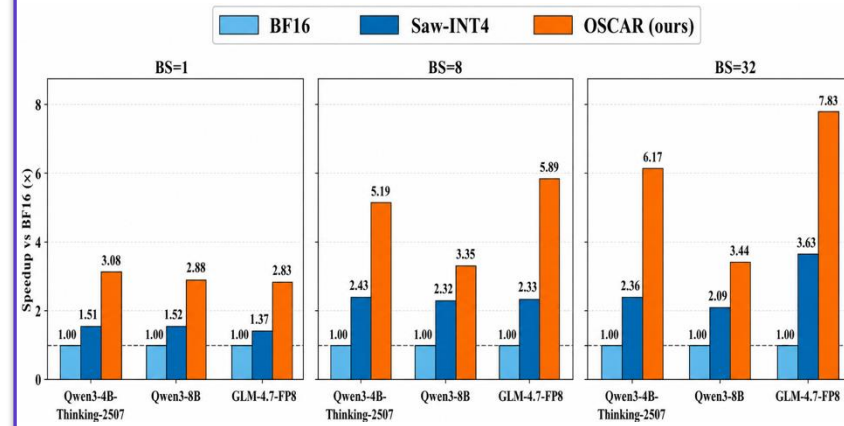
★ OSCAR: for read/write path—no extra memory traffic

Performance

Accuracy

Method	Qwen3-4B thinking-2507	Qwen3-8B	Qwen3-32B	GLM-4.7 FP8, 358B
BF16	75.64	70.84	74.19	77.89
Saw-INT4	73.11	69.97	74.43	77.95
TurboQuant	31.74	56.88	71.99	78.15
Naive INT2	0.00	0.00	0.00	60.49
QuaRot-INT2	1.40	10.14	7.90	75.14
OSCAR (ours)	71.86	69.42	74.17	78.16

Throughput



Amazing Performance

★ OSCAR: accuracy preserve and high throughput